

Flam Data Analyst Assignment – User & Campaign Insights

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Tools : Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn)

Datasets : users.csv, campaigns.csv, usage_metrics.csv

Section A – Exploratory Analysis

The datasets were cleaned and merged to explore user activity and campaign performance on Flam. **Duplicates were removed, missing values handled ('Unknown' for country/device, '0' for numeric fields), and an 'is_published' flag was created from campaign status. Cleaned CSV files were saved for reproducibility. Exploration revealed that mobile users showed higher activity, and countries like India and the USA had the largest active user base.**

Section B – Behavioral Insights

Behavioral trends were derived from merged user–campaign data. Campaigns using higher credits were more likely to be published, and referred users consistently published more campaigns. **Publishing activity correlated with credit investment and referral-based acquisition.**

Section C – Experience & Performance Patterns

Grouping by experience_type reveals that immersive formats such as **3D Experience and Spatial Interactive have the highest average credits used and publish rates, indicating a strong creator preference for richer, more engaging experiences.** In contrast, **less interactive formats like Prism Extended and Spatial Alpha show lower engagement, suggesting limited adoption.**

Section D – Visualization & Storytelling

Visualizations were created to communicate key insights effectively: **(1) Bar plot of Average Credits Used by Experience Type, (2) Heatmap of Average Credits Used by Country and Device, and (3) Confusion Matrix showing model accuracy.** These visuals highlight relationships between creator behavior, device usage, and publishing outcomes.(all outputs mentioned below).

Section E – Insights & Recommendations

Key Insights:

- **Higher credits and immersive experience types strongly increase publishing probability.**
- **Referral users are more active and publish more campaigns.**
- **Mobile creators dominate engagement and credit usage.**
- **Engagement is concentrated in a few top-performing countries.**

Recommendations:

- 1. Expand referral incentives to attract more active creators.**
- 2. Promote 3D Video experiences for higher engagement.**
- 3. Encourage credit usage with rewards or bundles.**
- 4. Optimize mobile creation tools for better user experience.**
- 5. Prioritize marketing and product support in high-engagement regions.**

Bonus – Predictive Modeling

A **Logistic Regression model** predicted campaign publishing using **credits_used**, **experience_type**, **country**, and **device_type**. Model accuracy reached **97%**, with precision of **1.00** and recall of **0.93**. **Support: Class 0 (Unpublished) → 73 samples (57%); Class 1 (Published) → 56 samples (43%)**. This confirms that publishing behavior is highly predictable, driven by creator investment and content type. The model showed **no signs of overfitting or underfitting** – train and test accuracies were consistent, indicating **strong generalization and reliable performance**.

Conclusion & Outputs

Flam’s data shows that campaign success depends on credits used, experience type, and creator engagement. By leveraging these insights and optimizing referrals, credits, and mobile experiences, Flam can enhance publishing rates and overall creator activity on the platform.

